

LOGIC No: SEQ-1201	DESCRIPTION: HEXANE FEED PUMP (GA-1201A/B) SHUTDOWN LOGIC	SIL: SIL 1	TAG: GA-1201A FLOC: TJC-LLD-1200-01
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CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 TRIP PUMP MOTOR GA-1201A	EFF-2 CLOSE DISCHARGE XV-1201	EFF-3 OPEN MIN-FLOW FV-1201	EFF-4 ANNUNCIATE ALARM ON DCS	EFF-5 START STANDBY PUMP GA-1201B (auto)
T1	Suction pressure LOW-LOW	PSLL-1201	< 0.5 barg	2oo3	X	X		X	X
T2	Discharge flow LOW-LOW (min flow)	FSLL-1201	< 9 m3/h for 30 s	1oo1	X		X	X	X
T3	Bearing vibration HIGH-HIGH	VSHH-1201	> 7.1 mm/s RMS	1oo2	X	X		X	X
T4	Bearing temperature HIGH-HIGH	TSHH-1201	> 95 degC	1oo1	X	X		X	
T5	Motor overload / electrical fault	MPR-1201	MPR relay pickup	1oo1	X	X		X	X
T6	Local/DCS emergency stop	HS-1201	Manual	1oo1	X	X		X	

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	TRIP PUMP MOTOR GA-1201A
EFF-2	CLOSE DISCHARGE XV-1201
EFF-3	OPEN MIN-FLOW FV-1201
EFF-4	ANNUNCIATE ALARM ON DCS
EFF-5	START STANDBY PUMP GA-1201B (auto)

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Suction valve OPEN	ZSO-1201
2	Seal flush established (dP > 1.5 bar)	PDI-1201
3	Min-flow valve OPEN	ZSO-1202
4	No active trip / reset done	DCS reset
=>	ALL permissives TRUE (AND) => START GA-1201A ENABLED	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID; TJC-LLD-PID-1201 for instrument loops and the Datasheet for design limits of GA-1201A.